

Frontend / Backend Integration Scan

本文件記錄目前前後端整合健檢結果，重點是確認前端可達頁面與按鈕是否能連到後端 API 與 PostgreSQL。

Current Summary

- Backend is connected to PostgreSQL through `backend/src/db.mjs`.
- Runtime API logic uses `backend/src/pgRepository.mjs`.
- Database setup is available through `npm run db:setup`.
- Frontend API client is `src/api/client.ts`.
- Frontend default API URL is `http://127.0.0.1:8787/api/v1`.
- Main reachable pages are wired to backend APIs.
- Build and lint are passing after the latest backend fix.

Current Progress

- ☒ Inventory frontend routes and clickable controls.
- ☒ Map frontend controls to backend API/data needs.
- ☒ Connect backend runtime to PostgreSQL.
- ☒ Add database setup command.
- ☒ Add frontend API client.
- ☒ Wire nurse overview to backend.
- ☒ Wire burden form buttons to backend persistence.
- ☒ Wire TO-DO checkbox updates to backend persistence.
- ☒ Wire allocation suggestion, reassignment, and confirmation to backend persistence.
- ☒ Wire allocation result to backend handoff sheet.
- ☒ Wire war-room to backend data.
- ☒ Run `npm run lint`.
- ☒ Run `npm run build`.
- ☐ Route and implement order import.
- ☐ Persist formal handoff snapshots.
- ☐ Add automated E2E tests.

Route And Button Status

AppShell Navigation

Status: working.

- 護理師首頁 : navigates to `/nurse/overview`.
- 麻煩度填寫 : navigates to `/nurse/burden-form`.
- T0-D0 : navigates to `/nurse/todo`.
- 指派分床配對 : navigates to `/leader/allocation`.
- 查看分床結果 : navigates to `/leader/allocation-result`.
- 戰情室 : navigates to `/leader/war-room`.

/nurse/overview

Status: backend-connected read-only page.

- Reads GET `/api/v1/nurse/overview` .
- Reads burden assessments for summary display.
- No write buttons on this page.

/nurse/burden-form

Status: backend-connected and persistent.

- 客觀 (系統) tab: reads objective values from backend.
- 主觀 (自填) tab: edits local form state before save.
- RASS input: supports number or blank.
- Yes/No buttons: update form state.
- Low/Mid/High buttons: update form state.
- Header 儲存 : persists submitted data with PATCH `/api/v1/burden-assessments/{assessmentId}` .
- Bottom 儲存草稿 : persists draft data.
- Bottom 送出 : persists submitted data.

Backend notes:

- Subjective null values now delete the corresponding `burden_values` row.
- This avoids `burden_values_check` violations when `rassScore` is blank.

/nurse/todo

Status: backend-connected and persistent.

- 全部, 待完成, 已完成 : local filter over backend task data.
- Kind chips: local filter over backend task data.
- 清除 : clears local kind filter.
- Task checkbox: calls PATCH `/api/v1/tasks/{taskId}` and persists status.
- +N 更多, 顯示全部, 收合 : local expand/collapse UI state.

/leader/allocation

Status: backend-connected and persistent.

- Initial run: reads seeded allocation run when available.
- 系統建議分床 : calls POST `/api/v1/allocation-runs/suggest` .
- Nurse reassignment select: calls PUT `/api/v1/allocation-runs/{allocationRunId}/items` .
- 確認送出 : calls POST `/api/v1/allocation-runs/{allocationRunId}/confirm` , then navigates to allocation result.

/leader/allocation-result

Status: backend-connected read-only page.

- Reads GET `/api/v1/handoff-sheets?shiftId={shiftId}` .

- Shows patient, bed, diagnosis, burden score, current nurse, and next nurse.

Current limitation:

- This is a live derived handoff sheet, not a persisted historical snapshot table yet.

/leader/war-room

Status: backend-connected read-only page with local expand/collapse.

- Reads `GET /api/v1/war-room?shiftId={shiftId}` .
- Uses backend allocation, burden score, and task data.
- 展開細節 / 收起細節 : local UI state only.
- Expanded checkboxes are read-only display.

OrdersImportPage

Status: not reachable.

- `src/pages/OrdersImportPage.tsx` exists.
- It is not mounted in `src/App.tsx` .
- Step 4 order import API is not implemented yet.

Backend API Coverage

Verified implemented endpoints:

- `GET /api/v1/health`
- `GET /api/v1/me`
- `GET /api/v1/shifts/current`
- `GET /api/v1/nurses?shiftId={shiftId}`
- `GET /api/v1/admissions?shiftId={shiftId}&status=active`
- `GET /api/v1/nurse/overview?shiftId={shiftId}`
- `GET /api/v1/burden-assessments?shiftId={shiftId}&scope=mine`
- `PATCH /api/v1/burden-assessments/{assessmentId}`
- `GET /api/v1/tasks?shiftId={shiftId}&assignee=me`
- `PATCH /api/v1/tasks/{taskId}`
- `POST /api/v1/allocation-runs/suggest`
- `GET /api/v1/allocation-runs/{allocationRunId}`
- `PUT /api/v1/allocation-runs/{allocationRunId}/items`
- `POST /api/v1/allocation-runs/{allocationRunId}/confirm`
- `GET /api/v1/war-room?shiftId={shiftId}`
- `GET /api/v1/handoff-sheets?shiftId={shiftId}`

Database Coverage

Database artifacts present:

- `backend/db/migrations/001_core_schema.sql`
- `backend/db/migrations/002_burden_tasks_schema.sql`
- `backend/db/migrations/003_allocation_schema.sql`

- `backend/db/seeds/001_demo_core_data.sql`
- `backend/db/seeds/002_demo_burden_tasks_data.sql`
- `backend/db/seeds/003_demo_allocation_data.sql`

Runtime DB wiring:

- `pg` dependency installed.
- `DATABASE_URL` default: `postgresql://postgres@%2Ftmp/sad_frontend_v2` .
- `DATABASE_ADMIN_URL` default: `postgresql://postgres@%2Ftmp/postgres` .
- `npm run db:setup` resets and seeds the local DB.

Verification Log

Commands run successfully:

```
npm run lint
npm run build
```

Recent backend issue fixed:

- Error: `new row for relation "burden_values" violates check constraint "burden_values_check"` .
- Cause: blank `rassScore` sent `null` , but backend attempted to insert a `burden_values` row with all value columns null.
- Fix: `upsertSubjectiveValue` now deletes the value row when the incoming value is `null` .

Previously verified API smoke flow:

- `GET /health` : ok.
- `GET /nurse/overview` : returned current nurse patients.
- `PATCH /burden-assessments/{id}` : saved subjective values.
- `PATCH /tasks/{id}` : toggled task status.
- `POST /allocation-runs/suggest` : created DB-backed draft run.
- `POST /allocation-runs/{id}/confirm` : confirmed run with no unassigned patients.
- `GET /war-room` : returned nurse/task data.
- `GET /handoff-sheets` : returned handoff rows.

How To Run Locally

Backend:

```
npm run db:setup
npm run api:dev
```

Frontend:

```
npm run dev
```

If frontend shows `Failed to fetch` :

1. Confirm backend is running.
2. Check `curl http://127.0.0.1:8787/api/v1/health` .
3. If backend uses another port, set `VITE_API_BASE_URL` .

Remaining Risks / Gaps

- Step 4 order import is not implemented.
- `OrdersImportPage.tsx` is not routed.
- Handoff sheet is derived live, not persisted as formal snapshot rows.
- Auth is demo-only via default user or `X-User-Id` .
- No automated E2E tests yet.